HARNESSING DATA CENTRE EXPERTISE TO DRIVE FORWARD INSTITUTIONAL RESEARCH DATA MANAGEMENT

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PROJECT BACKGROUND

- Rapid evolution of research data management at UK higher education institutions
- A number of projects in the UK over the past few years as part of Jisc's Managing Research Data Programme

- Research Data @Essex project at University of Essex, as part infrastructure strand, Nov 2011 - March 2013
- UK Data Archive lead, in collaboration with University of Essex IT & research support

UK Data Archive & University of Essex

- UK Data Archive is curator of the largest collection of social science data in the UK
  - 1000's of datasets used by researchers, teachers, policy makers etc.
  - Manage the UK Data Service and are experts in issues surrounding data management and sharing

- Is based at University of Essex
  - Recognised internationally for quality of research
  - 18 schools/departments, spanning Humanities, Law, Business, Social Sciences, Biology, Engineering +

WHAT DO IR'S AND DATA CENTRES **HAVE IN COMMON?**

Both attempt to enable:

- **Storage** – somewhere researchers can drop off their data *after* research and (in theory) forget about it
- **Re-use** – provide documentation and comprehensive descriptive metadata
- **Access** – download services for data in storage, preferably at a granular (file) level
- **Discovery** – some kind of search/browse functionality using appropriate catalogue metadata

WHAT DO IR'S AND DATA CENTRES **DO DIFFERENTLY?**

Data centres will provide a higher level of end user support than IR's, simply for resource reasons.

IR's don't care about:

- **Curation** – stuff goes in with minimal editorial control, and are probably not touched again
- **Tech support** – likely to be a single contact; not a service = important that it's easy to use
- **Data quality** – no file checking or any such validation of datasets

OUR APPROACH

- Piloting a research data management infrastructure at Essex
- Reusable outputs that could meet the needs of both Essex and in theory any university in UK
- Interaction with researchers at Essex throughout development, to ensure outputs aligned with their needs

- Two key technical outputs:
  1. **Metadata profile (mapped to components)**
  2. **ReCollect research data plugin for EPrints**

METADATA PROFILE

- Define a set of generic metadata elements to allow the effective description of research data

Core
- Metadata for **findability**
- e.g. author, discipline, project

Detail
- **Generic** dataset **description**
- e.g. collection methodology

Discipline
- As additional **file**
- Structured e.g. XML

Based on JISC IDMB project outputs, Southampton University

METADATA PROFILE

- Uptake within MRD community of DataCite core as minimal metadata standard
- But no consensus as to further detail

- What would the requirements be for this detail?
  - Generic and flexible
  - Standards compliant

- Minimising barriers for researchers to deposit data
  - …while satisfying requirements for re-use

METADATA PROFILE

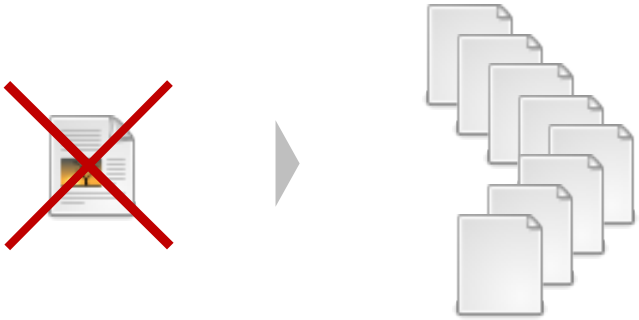
- We took standards based approach, producing:
  1. **Metdata mapping** between relevant schema
  2. **Metadata profile** built from mapped elements

- Standards that we mapped to, and form profile components:
  - **DataCite** – minimal mandatory metadata (discovery)
  - **INSPIRE** – basic descriptive detail
  - **DDI 2.1** – further descriptive detail e.g. collection methodology

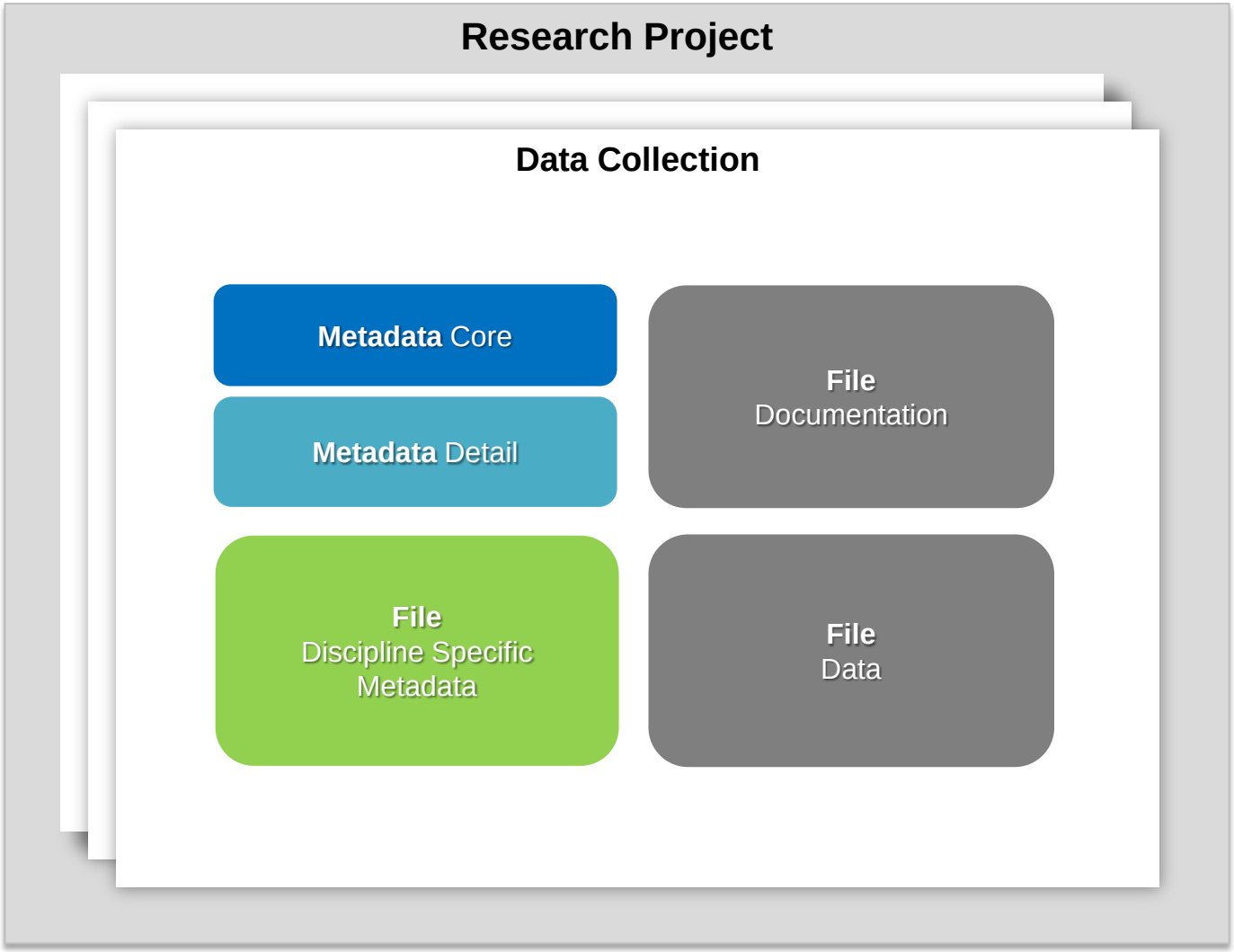
REPOSITORY PILOT

- EPrints is a widely used, open source institutional repository solution
  - Geared toward article type deposits

- Development work at Essex to adapt for data
  - Complex and varied data collections from a wide range of disciplines



OBJECT MODEL



REPOSITORY PILOT

- Pilot institutional data repository at Essex the first of its kind for EPrints
- EPrints already has great features – e.g. intuitive deposit workflow; embargo and restrict data
- But extensive customisation needed, including:
  - Integrating the metadata profile previously described
  - Conforming to object model previously described
  - Customisation of the user interface to accommodate complex collections
  - Tweaking the workflow and updating user documentation

Sample Data: Bioimaging - Surface Receptors

Laissue, Philippe (2012) *Sample Data: Bioimaging - Surface Receptors*. [Data Collection]. Research Data Essex, University of Essex. http://researchdata.essex.ac.uk/id/eprint/77 (Unpublished)

Collection description

PLEASE NOTE THAT THIS IS A SAMPLE DATA COLLECTION MADE VISIBLE FOR DEMONSTRATION PURPOSES ONLY. Confocal microscopy derived images of cells, used to demonstrate the development of a novel method for the quantified, superresolved, three-dimensional colocalisation of histocompatibility class I receptors.

Keywords:	Colocalisation, quantification, confocal microscopy, surface receptors, bioimage informatics
Subjects:	Biological Sciences > Biology > Cell Biology
Divisions:	Faculty of Science and Engineering > Biological Sciences, School of
Depositing User:	RD Admin
Date Deposited:	22 Aug 2012 10:58
Revision:	78
URI:	http://researchdata.essex.ac.uk/id/eprint/77

Additional details

Available Files

Data

data_tiff.zip

Visible to:	Repository staff only
Content:	Data
Document description:	Images (TIFF format)
Metadata Revision:	5
Mime-Type:	application/zip
File size:	2175Mb

data_nd2.zip

Documentation

doc_thumbnails.zip

Time period

Collection period: The time period (start and end date) for collection of the data, e.g. dates of fieldwork.

From — Year: Month: Unspecified Day: ? To — Year: Month: Unspecified Day: ?

Temporal coverage: From — Year: Month: Unspecified Day: ? To — Year: Month: Unspecified Day: ?

Geographic coverage

Geographic location

Data collection method

Legal and ethical issues

Data processing and preparation activities

Resource language

English

Additional information

Related resources

URL	Type
	UNSPECIFIED

More input rows

RECOLLECT

ReCollect

RESEARCH DATA PLUGIN FOR EPRINTS

- EPrints tailored for research data
- Enhanced presentation of data collections
- Standards based descriptive metadata
- One click install via Bazaar 'app store'

bazaar.eprints.org
or install via EPrints admin interface

ADDITIONAL MATERIALS

- Supporting project outputs include:
  - **Repository user documentation** i.e. how to deposit, esp metadata guidance
  - **Essex data repository policies** such as deposit terms & conditions
  - **Detailed report on repository development** including design rationale and user testing

- All published via project page on UKDA website

LOOKING FORWARD

- ReCollect model is reusable, adaptable
  - Potential uptake by many institutions
  - Including customised instances e.g. Southampton have integrated with generic digital repository
- Collaborating with the above for wider ‘joined-up-ness’ of approaches – will put forward a community recommendation
- Metadata profile also being adopted
  - Proposes a very flexible set of elements
  - A work to be built upon / refined

CONTACT

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